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h) Hypergeometric Distribution

number of success (x) in n draws, without replacement

from a finite population N , that contains K successes

$$X \sim \text{HG}(N, K, n)$$

$$P_X(x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}$$

$$, x \in \mathbb{N}, \text{ for } P_X(x) > 0$$

$$x \leq K, n-x \leq N-K \rightarrow n+K-N \leq x$$

$$x \leq \min(K, n), x \geq \max(0, n+K-N)$$

Exercise

$$\sum_{x=0}^n \binom{K}{x} \binom{N-K}{n-x} = \binom{N}{n}$$

$$\Rightarrow \sum_{x=0}^n P_X(x) = 1$$

when $N \rightarrow \infty$, the effect of sampling without replacement becomes less and less
 $\Rightarrow$ success probability K/N and it can be modeled as binomial distribution
with $n, p = K/N$

$$X \sim \text{Bin}(n, K/N)$$

$$X_N \sim \text{HG}(N, pN, n), X \sim \text{Bin}(n, p)$$

$$\lim_{N \rightarrow \infty} P(X_N = x) = P(X = x)$$

i) Multinomial Distribution

$$P(x_1, x_2, x_3, \dots, x_r) = \frac{n!}{x_1! x_2! \dots x_r!} p_1^{x_1} p_2^{x_2} p_3^{x_3} \dots p_r^{x_r}$$

$$\text{where } \sum_{i=1}^r x_i = n, \sum_{i=1}^r p_i = 1$$

Multiple Random Variables

Joint pmfs: $P_{X,Y}: \mathbb{R}^2 \rightarrow [0,1]$

$$P_{X,Y}(x,y) = P(X=x, Y=y) = P(\{\omega \mid X(\omega)=x\} \cap \{\omega \mid Y(\omega)=y\}) \quad , \forall x,y \in \mathbb{R}, (x,y) \in \mathbb{R}^2$$

if $X_1, X_2, \dots, X_n$ are random variables

$$P_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = P(X_1=x_1, X_2=x_2, \dots, X_n=x_n)$$

eg) 100 independent coin tosses

define X : number of head, Y : number of tail

$$P_{X,Y}(1,1) = P(X=1, Y=1) = 0$$

$$P_{X,Y}(1,99) = P(X=1, Y=99) = \binom{100}{1} (1/2)^1 \cdot (1/2)^{99} = \frac{100}{2^{100}}$$

Marginal pmfs

$$P_X(x) = P(X=x) = \sum_{y'} P_{X,Y}(x, y')$$

$$P_Y(y) = P(Y=y) = \sum_{x'} P_{X,Y}(x', y)$$

The Law of Total Probability

$$P_X(x) = P(X=x) = P(\{X=x\} \cap \Omega)$$

$$B_i = \{Y=y_i\}, \quad B_i \cap B_j = \emptyset \quad \forall i \neq j \\ \bigcup_{i \in \mathbb{N}} B_i = \Omega$$

$$\Rightarrow P_X(x) = P(\{X=x\} \cap \bigcup_{i \in \mathbb{N}} B_i)$$

$$= P(\bigcup_{i \in \mathbb{N}} \{X=x\} \cap \{Y=y_i\}) = \sum_{i \in \mathbb{N}} P(X=x, Y=y_i) = \sum_{y'} P_{X,Y}(x, y')$$

Conditional pmfs

$$P_{X|Y}(x|y) = \frac{P(X=x, Y=y)}{P(Y=y)} = \frac{P_{XY}(x,y)}{P_Y(y)}, \text{ given } P_Y(y) > 0$$

$$P_{X_1, X_2, \dots, X_n | Y_1, Y_2, \dots, Y_m}(x_1, x_2, \dots, x_n | y_1, y_2, \dots, y_m) = \frac{P_{X_1, X_2, \dots, X_n, Y_1, Y_2, \dots, Y_m}(x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_m)}{P_{Y_1, Y_2, \dots, Y_m}(y_1, y_2, \dots, y_m)}$$

$$\text{eg.) } P_{XY}(x,y) = \begin{cases} \frac{2}{n(n+1)} \left(\frac{x}{n+1}\right)^y & x=0, 1, \dots, n-1, y=0, 1, 2, \dots \\ 0 & \text{otherwise} \end{cases}$$

Find $P_{Y|X}(y|x)$

$$P_{Y|X}(y|x) = \frac{P_{XY}(x,y)}{P_X(x)} \quad \frac{2}{n(n+1)} \frac{1}{1 - \frac{x}{n+1}} = \frac{2}{n(n+1)} (1+x)$$

$$P_X(x) = \begin{cases} \sum_{y=0}^{\infty} \frac{2}{n(n+1)} \left(\frac{x}{n+1}\right)^y & y \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$P_{Y|X}(y|x) = \begin{cases} \left(\frac{x}{n+1}\right)^y \frac{1}{1-x} & x=0, 1, \dots, n-1, y=0, 1, 2, \dots \\ 0 & \text{otherwise} \end{cases}$$

$$Y|X=x \sim \text{Geo.} \left(\frac{1}{1+x} \right)$$

Exercise)

$$P_{X|N}(x|n) = \frac{1}{(n+1-1)+1} = \frac{1}{n+1}$$

$$P_N(n) = \frac{e^{-\lambda} \lambda^n}{n!} \quad n=0, 1, 2, \dots$$

Find pmf of X

$$P_{X,N}(x,n) = P_{X|N}(x|n) \cdot P_N(n) = \begin{cases} \frac{1}{n+1} \cdot \frac{e^{-\lambda} \lambda^n}{n!} & x=1, 2, \dots, n+1 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} P_X(x) &= \sum_{n=0}^{\infty} P_{X,N}(x,n) = \sum_{n=x-1}^{\infty} \frac{e^{-\lambda} \lambda^n}{(n+1)!} \\ &= \sum_{t=x}^{\infty} \frac{e^{-\lambda} \lambda^t}{t!} = \frac{1}{\lambda} e^{-\lambda} \left(\sum_{t=0}^{\infty} \frac{\lambda^t}{t!} - \sum_{t=0}^{x-1} \frac{\lambda^t}{t!} \right) \\ &= \frac{1}{\lambda} e^{-\lambda} \left(e^{\lambda} - \sum_{t=0}^{x-1} \frac{\lambda^t}{t!} \right) \end{aligned}$$

Law of total probability $P(Y=y) = \sum_i P(Y=y | X=x_i) P(X=x_i)$

Baye's Rule $P(Y=y | X=x) = \frac{P(X=x | Y=y) P(Y=y)}{\sum_y P(X=x | Y=y) P(Y=y)}$

The Substitution Law : X, Y are discrete r.v. on $(\Omega, \mathcal{F}, P)$ and $Z = g(X, Y)$
where $Z: \mathbb{R} \rightarrow \mathbb{R}$

$$P(Z=z) = P(g(X, Y)=z) \\ = \sum_i P(g(X, Y)=z | X=x_i) P(X=x_i)$$

Independence of Discrete Random Variables

b) $\forall x, y \in \mathbb{R}$ the events $\{\omega | X(\omega)=x\}$ and $\{\omega | Y(\omega)=y\}$ are independent

c) $\forall x, y \in \mathbb{R} \quad P_{X,Y}(x, y) = P_X(x) P_Y(y)$

d) $\forall x, y \in \mathbb{R}$ if $P_Y(y) > 0$, $P_{X,Y}(x, y) = P_X(x)$

Exercise (c) $\Rightarrow$ (a)

$$B_1, B_2 \in \mathcal{B}(\mathbb{R})$$

$$P(X \in B_1, Y \in B_2) = P(X \in B_1) P(Y \in B_2)$$

based on assumption that $P_{X,Y}(x, y) = P_X(x) P_Y(y)$

$$\begin{aligned} \Rightarrow P(X \in B_1, Y \in B_2) &= \sum_{\substack{x \in B_1 \\ y \in B_2}} P_{X,Y}(x, y) \\ &= \sum_{x \in B_1} P_X(x) \sum_{y \in B_2} P_Y(y) \\ &= P(X \in B_1) P(Y \in B_2) \end{aligned}$$

X, Y are independent discrete random variables on $(\Omega, \mathcal{F}, P)$
 g, h real function $g: \mathbb{R} \rightarrow \mathbb{R}, h: \mathbb{R} \rightarrow \mathbb{R}$

then $g(X), h(Y)$ are independent

Derived Distributions (Function of Discrete R.V's)

given discrete random variable with density P_X , and function $g: \mathbb{R} \rightarrow \mathbb{R}$.
find CDF/PMF of the random variables $Y = g(X)$

$$P(Y=y) = P_Y(y) = \sum_{\{x|y=g(x)\}} P_X(x)$$